dec

March 10, 2026

[1]:
```
from keras.datasets import mnist
import numpy as np
np.random.seed(10)
```

```
2026-03-10 09:45:20.251431: E
external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:477] Unable to register
cuFFT factory: Attempting to register factory for plugin cuFFT when one has
already been registered
WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
E0000 00:00:1773132320.264697  125918 cuda_dnn.cc:8310] Unable to register cuDNN
factory: Attempting to register factory for plugin cuDNN when one has already
been registered
E0000 00:00:1773132320.268966  125918 cuda_blas.cc:1418] Unable to register
cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has
already been registered
2026-03-10 09:45:20.283468: I tensorflow/core/platform/cpu_feature_guard.cc:210]
This TensorFlow binary is optimized to use available CPU instructions in
performance-critical operations.
To enable the following instructions: AVX2 FMA, in other operations, rebuild
TensorFlow with the appropriate compiler flags.
```

[2]:
```
from time import time
#import numpy as np
import tensorflow.keras.backend as K
from tensorflow.keras.layers import Layer, InputSpec
from keras.layers import Dense, Input
from keras.models import Model
from keras.optimizers import SGD
from keras import callbacks
from keras.initializers import VarianceScaling
from sklearn.cluster import KMeans
from sklearn import metrics
from keras.datasets import fashion_mnist
```

[3]:
```
def autoencoder(dims, act='relu', init='glorot_uniform'):
    """
    Fully connected auto-encoder model, symmetric.
```

```
    Arguments:
        dims: list of number of units in each layer of encoder. dims[0] is␣
↪input dim, dims[-1] is units in hidden layer.
            The decoder is symmetric with encoder. So number of layers of the␣
↪auto-encoder is 2*len(dims)-1
        act: activation, not applied to Input, Hidden and Output layers
    return:
        (ae_model, encoder_model), Model of autoencoder and model of encoder
    """
    n_stacks = len(dims) - 1
    # input
    input_img = Input(shape=(dims[0],), name='input')
    x = input_img
    # internal layers in encoder
    for i in range(n_stacks-1):
        x = Dense(dims[i + 1], activation=act, kernel_initializer=init,␣
↪name='encoder_%d' % i)(x)

    # hidden layer
    encoded = Dense(dims[-1], kernel_initializer=init, name='encoder_%d' %␣
↪(n_stacks - 1))(x)  # hidden layer, features are extracted from here

    x = encoded
    # internal layers in decoder
    for i in range(n_stacks-1, 0, -1):
        x = Dense(dims[i], activation=act, kernel_initializer=init,␣
↪name='decoder_%d' % i)(x)

    # output
    x = Dense(dims[0], kernel_initializer=init, name='decoder_0')(x)
    decoded = x
    return Model(inputs=input_img, outputs=decoded, name='AE'),␣
↪Model(inputs=input_img, outputs=encoded, name='encoder')
```

[4]:
```
#(x_train, y_train), (x_test, y_test) = mnist.load_data()
(x_train, y_train), (x_test, y_test) = fashion_mnist.load_data()


x = np.concatenate((x_train, x_test))
y = np.concatenate((y_train, y_test))
x = x.reshape((x.shape[0], -1))
x = np.divide(x, 255.)
```

[5]:
```
n_clusters = len(np.unique(y))
x.shape
```

[5]: (70000, 784)

[6]:
```
kmeans = KMeans(n_clusters=n_clusters, n_init=1)
y_pred_kmeans = kmeans.fit_predict(x)
```

[7]:
```
metrics.accuracy_score(y, y_pred_kmeans)
```

[7]: 0.07027142857142857

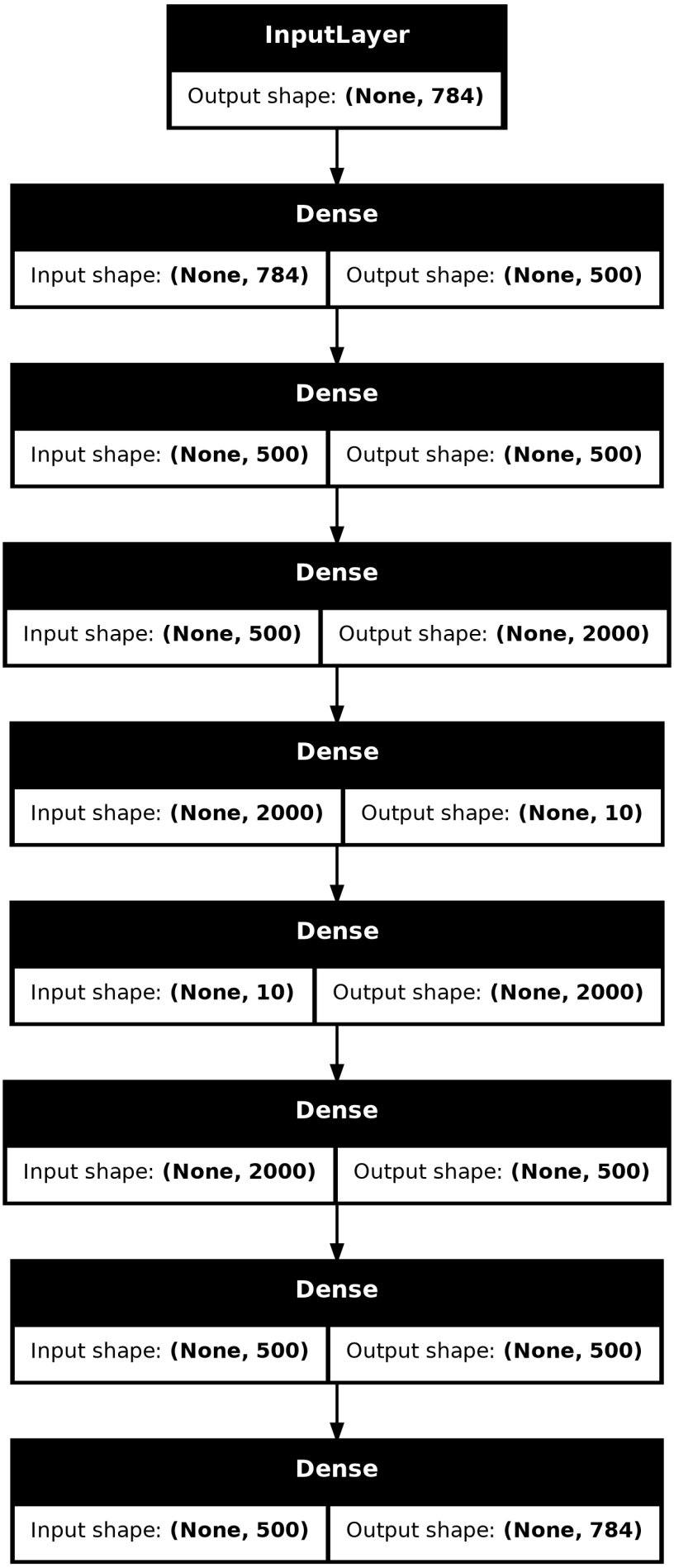
[8]:
```
dims = [x.shape[-1], 500, 500, 2000, 10]
init = VarianceScaling(scale=1. / 3., mode='fan_in',
                           distribution='uniform')
pretrain_optimizer = SGD(learning_rate=1.0, momentum=0.9)
pretrain_epochs = 30
batch_size = 256
save_dir = './results'
```

```
I0000 00:00:1773132327.555267  125918 gpu_device.cc:2022] Created device
/job:localhost/replica:0/task:0/device:GPU:0 with 22314 MB memory:  -> device:
0, name: NVIDIA GeForce RTX 3090, pci bus id: 0000:01:00.0, compute capability:
8.6
```

[9]:
```
autoencoder, encoder = autoencoder(dims, init=init)
```

[10]:
```
from keras.utils import plot_model
plot_model(autoencoder, to_file='autoencoder.png', show_shapes=True)
from IPython.display import Image
Image(filename='autoencoder.png')
```

[10]:


InputLayer
Output shape: (None, 784)
Dense
Input shape: (None, 784)
Output shape: (None, 500)
Dense
Input shape: (None, 500)
Output shape: (None, 500)
Dense
Input shape: (None, 500)
Output shape: (None, 2000)
Dense
Input shape: (None, 2000)
Output shape: (None, 10)
Dense
Input shape: (None, 10)
Output shape: (None, 2000)
Dense
Input shape: (None, 2000)
Output shape: (None, 500)
Dense
Input shape: (None, 500)
Output shape: (None, 500)
Dense
Input shape: (None, 500)
Output shape: (None, 784)

[11]:
```
autoencoder.compile(optimizer=pretrain_optimizer, loss='mse')
autoencoder.fit(x, x, batch_size=batch_size, epochs=pretrain_epochs) #,␣
↪callbacks=cb)
#autoencoder.save_weights(save_dir + '/ae_weights.h5')
```

```
Epoch 1/30

WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
I0000 00:00:1773132329.475526  126017 service.cc:148] XLA service 0x71ea74004830
initialized for platform CUDA (this does not guarantee that XLA will be used).
Devices:
I0000 00:00:1773132329.475562  126017 service.cc:156]   StreamExecutor device
(0): NVIDIA GeForce RTX 3090, Compute Capability 8.6
2026-03-10 09:45:29.489116: I
tensorflow/compiler/mlir/tensorflow/utils/dump_mlir_util.cc:268] disabling MLIR
crash reproducer, set env var `MLIR_CRASH_REPRODUCER_DIRECTORY` to enable.
I0000 00:00:1773132329.553189  126017 cuda_dnn.cc:529] Loaded cuDNN version
90800
2026-03-10 09:45:30.139515: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_48', 32 bytes spill stores, 32 bytes spill loads

2026-03-10 09:45:30.336697: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_62', 100 bytes spill stores, 100 bytes spill loads

2026-03-10 09:45:30.348316: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_48_0', 776 bytes spill stores, 724 bytes spill loads

2026-03-10 09:45:30.390239: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_48', 276 bytes spill stores, 280 bytes spill loads

2026-03-10 09:45:30.411297: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_55', 24 bytes spill stores, 28 bytes spill loads

2026-03-10 09:45:30.481099: I
```

```
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_55', 32 bytes spill stores, 32 bytes spill loads

2026-03-10 09:45:30.683613: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_48', 24 bytes spill stores, 28 bytes spill loads

2026-03-10 09:45:31.013141: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_55_0', 776 bytes spill stores, 724 bytes spill loads

2026-03-10 09:45:31.088189: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_55', 276 bytes spill stores, 280 bytes spill loads

2026-03-10 09:45:31.096361: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73', 44 bytes spill stores, 44 bytes spill loads

2026-03-10 09:45:31.201460: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_62', 440 bytes spill stores, 440 bytes spill loads

2026-03-10 09:45:31.460905: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_62', 204 bytes spill stores, 204 bytes spill loads

2026-03-10 09:45:31.618067: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73_0', 276 bytes spill stores, 276 bytes spill loads

2026-03-10 09:45:31.881048: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73', 20 bytes spill stores, 28 bytes spill loads

2026-03-10 09:45:32.011947: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246', 148 bytes spill stores, 148 bytes spill loads
```

```
2026-03-10 09:45:32.066803: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73', 276 bytes spill stores, 280 bytes spill loads

2026-03-10 09:45:32.085446: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73_0', 784 bytes spill stores, 736 bytes spill loads

2026-03-10 09:45:32.210433: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246_0', 60 bytes spill stores, 60 bytes spill loads

2026-03-10 09:45:32.294261: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_55', 260 bytes spill stores, 260 bytes spill loads

2026-03-10 09:45:32.540950: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_87_0', 776 bytes spill stores, 724 bytes spill loads

2026-03-10 09:45:32.633607: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_87', 24 bytes spill stores, 28 bytes spill loads

2026-03-10 09:45:32.848107: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246_0', 892 bytes spill stores, 612 bytes spill loads

2026-03-10 09:45:32.915016: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_87', 276 bytes spill stores, 280 bytes spill loads

2026-03-10 09:45:32.970698: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73', 212 bytes spill stores, 212 bytes spill loads

2026-03-10 09:45:33.100505: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
```

```
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_328_0', 88 bytes spill stores, 88 bytes spill loads

2026-03-10 09:45:33.274320: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_326', 44 bytes spill stores, 44 bytes spill loads

2026-03-10 09:45:33.356748: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246', 92 bytes spill stores, 92 bytes spill loads

2026-03-10 09:45:33.538156: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_326_0', 88 bytes spill stores, 88 bytes spill loads

2026-03-10 09:45:33.569700: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73', 48 bytes spill stores, 48 bytes spill loads

2026-03-10 09:45:33.726569: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_330_0', 88 bytes spill stores, 88 bytes spill loads

2026-03-10 09:45:33.735719: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_87', 32 bytes spill stores, 32 bytes spill loads

2026-03-10 09:45:33.842234: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246', 492 bytes spill stores, 492 bytes spill loads

2026-03-10 09:45:34.190397: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73', 16 bytes spill stores, 16 bytes spill loads

2026-03-10 09:45:34.285522: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246', 116 bytes spill stores, 116 bytes spill loads
```

```
2026-03-10 09:45:34.836653: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_330', 4 bytes spill stores, 4 bytes spill loads

2026-03-10 09:45:35.022651: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_340', 52 bytes spill stores, 52 bytes spill loads

2026-03-10 09:45:35.042786: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_336', 4 bytes spill stores, 4 bytes spill loads

2026-03-10 09:45:35.186928: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246', 16 bytes spill stores, 16 bytes spill loads

2026-03-10 09:45:35.524386: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_336_0', 88 bytes spill stores, 88 bytes spill loads

2026-03-10 09:45:36.017231: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_340_0', 228 bytes spill stores, 228 bytes spill loads


 79/274            0s 2ms/step -
loss: 0.1407

I0000 00:00:1773132337.947279  126017 device_compiler.h:188] Compiled cluster
using XLA!  This line is logged at most once for the lifetime of the process.

272/274            0s 2ms/step -
loss: 0.0990

2026-03-10 09:45:38.935445: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_48', 24 bytes spill stores, 24 bytes spill loads

2026-03-10 09:45:39.137743: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_48', 80 bytes spill stores, 80 bytes spill loads
```

```
2026-03-10 09:45:39.520638: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_55', 24 bytes spill stores, 24 bytes spill loads

2026-03-10 09:45:39.555148: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_48_0', 1044 bytes spill stores, 732 bytes spill loads

2026-03-10 09:45:39.843707: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_55_0', 1044 bytes spill stores, 732 bytes spill loads

2026-03-10 09:45:39.905140: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_62', 524 bytes spill stores, 416 bytes spill loads

2026-03-10 09:45:40.058647: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_55', 80 bytes spill stores, 80 bytes spill loads

2026-03-10 09:45:40.269601: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_48', 932 bytes spill stores, 764 bytes spill loads

2026-03-10 09:45:40.282057: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73', 28 bytes spill stores, 28 bytes spill loads

2026-03-10 09:45:40.424874: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_55', 60 bytes spill stores, 60 bytes spill loads

2026-03-10 09:45:40.528988: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_48', 60 bytes spill stores, 60 bytes spill loads

2026-03-10 09:45:40.736477: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
```

```
'gemm_fusion_dot_73', 88 bytes spill stores, 92 bytes spill loads

2026-03-10 09:45:40.751267: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_87', 932 bytes spill stores, 764 bytes spill loads

2026-03-10 09:45:40.917481: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_87', 24 bytes spill stores, 24 bytes spill loads

2026-03-10 09:45:40.971633: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_62', 212 bytes spill stores, 212 bytes spill loads

2026-03-10 09:45:41.039125: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_87', 60 bytes spill stores, 60 bytes spill loads

2026-03-10 09:45:41.045138: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246', 128 bytes spill stores, 128 bytes spill loads

2026-03-10 09:45:41.081243: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73_0', 1060 bytes spill stores, 748 bytes spill loads

2026-03-10 09:45:41.097756: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73', 980 bytes spill stores, 832 bytes spill loads

2026-03-10 09:45:41.117890: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_87', 80 bytes spill stores, 80 bytes spill loads

2026-03-10 09:45:41.374331: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_55', 260 bytes spill stores, 260 bytes spill loads

2026-03-10 09:45:41.414945: I
```

```
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246_0', 900 bytes spill stores, 616 bytes spill loads

2026-03-10 09:45:41.571252: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73', 28 bytes spill stores, 28 bytes spill loads

2026-03-10 09:45:41.667906: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_55', 932 bytes spill stores, 764 bytes spill loads

2026-03-10 09:45:41.717053: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_326', 4 bytes spill stores, 4 bytes spill loads

2026-03-10 09:45:41.796220: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_328', 12 bytes spill stores, 12 bytes spill loads

2026-03-10 09:45:41.917772: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73_0', 264 bytes spill stores, 264 bytes spill loads

2026-03-10 09:45:41.920349: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246', 520 bytes spill stores, 524 bytes spill loads

2026-03-10 09:45:42.219806: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_62', 100 bytes spill stores, 100 bytes spill loads

2026-03-10 09:45:42.293797: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246', 1484 bytes spill stores, 1080 bytes spill loads

2026-03-10 09:45:42.328673: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73', 256 bytes spill stores, 256 bytes spill loads
```

```
2026-03-10 09:45:42.468861: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_326', 84 bytes spill stores, 84 bytes spill loads

2026-03-10 09:45:42.489963: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_330', 68 bytes spill stores, 68 bytes spill loads

2026-03-10 09:45:42.875430: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_326', 8 bytes spill stores, 8 bytes spill loads

2026-03-10 09:45:43.092587: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_328', 80 bytes spill stores, 80 bytes spill loads

2026-03-10 09:45:43.287585: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_326', 36 bytes spill stores, 36 bytes spill loads

2026-03-10 09:45:43.458928: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_330', 60 bytes spill stores, 60 bytes spill loads

2026-03-10 09:45:43.506210: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_246', 92 bytes spill stores, 92 bytes spill loads

2026-03-10 09:45:43.516237: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_73', 28 bytes spill stores, 28 bytes spill loads

2026-03-10 09:45:43.574509: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_330', 12 bytes spill stores, 12 bytes spill loads

2026-03-10 09:45:43.603788: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
```

```
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_336', 12 bytes spill stores, 12 bytes spill loads

2026-03-10 09:45:44.528545: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_340', 96 bytes spill stores, 96 bytes spill loads

2026-03-10 09:45:44.529832: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_87_0', 1044 bytes spill stores, 732 bytes spill loads

2026-03-10 09:45:44.641661: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_336', 60 bytes spill stores, 60 bytes spill loads

2026-03-10 09:45:44.708775: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_340', 4 bytes spill stores, 4 bytes spill loads

2026-03-10 09:45:44.784099: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_340', 24 bytes spill stores, 24 bytes spill loads

2026-03-10 09:45:44.921442: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_328', 68 bytes spill stores, 68 bytes spill loads

2026-03-10 09:45:44.975536: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_340', 168 bytes spill stores, 180 bytes spill loads

2026-03-10 09:45:45.254298: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_340', 80 bytes spill stores, 100 bytes spill loads

2026-03-10 09:45:45.406758: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_336', 68 bytes spill stores, 68 bytes spill loads
```

```
274/274            19s 36ms/step -
loss: 0.0986
Epoch 2/30
274/274            1s 2ms/step -
loss: 0.0349
Epoch 3/30
274/274            1s 2ms/step -
loss: 0.0264
Epoch 4/30
274/274            1s 2ms/step -
loss: 0.0234
Epoch 5/30
274/274            1s 2ms/step -
loss: 0.0218
Epoch 6/30
274/274            1s 2ms/step -
loss: 0.0201
Epoch 7/30
274/274            1s 2ms/step -
loss: 0.0189
Epoch 8/30
274/274            1s 2ms/step -
loss: 0.0181
Epoch 9/30
274/274            1s 2ms/step -
loss: 0.0174
Epoch 10/30
274/274            1s 2ms/step -
loss: 0.0169
Epoch 11/30
274/274            1s 2ms/step -
loss: 0.0164
Epoch 12/30
274/274            1s 2ms/step -
loss: 0.0161
Epoch 13/30
274/274            1s 2ms/step -
loss: 0.0156
Epoch 14/30
274/274            1s 2ms/step -
loss: 0.0153
Epoch 15/30
274/274            1s 2ms/step -
loss: 0.0151
Epoch 16/30
274/274            1s 2ms/step -
loss: 0.0148
Epoch 17/30
```

```
274/274              1s 2ms/step -
loss: 0.0146
Epoch 18/30
274/274              1s 2ms/step -
loss: 0.0145
Epoch 19/30
274/274              1s 2ms/step -
loss: 0.0142
Epoch 20/30
274/274              1s 2ms/step -
loss: 0.0142
Epoch 21/30
274/274              1s 2ms/step -
loss: 0.0140
Epoch 22/30
274/274              1s 2ms/step -
loss: 0.0138
Epoch 23/30
274/274              1s 2ms/step -
loss: 0.0137
Epoch 24/30
274/274              1s 2ms/step -
loss: 0.0136
Epoch 25/30
274/274              1s 2ms/step -
loss: 0.0135
Epoch 26/30
274/274              1s 2ms/step -
loss: 0.0134
Epoch 27/30
274/274              1s 2ms/step -
loss: 0.0133
Epoch 28/30
274/274              1s 2ms/step -
loss: 0.0133
Epoch 29/30
274/274              1s 2ms/step -
loss: 0.0131
Epoch 30/30
274/274              1s 2ms/step -
loss: 0.0131
```

[11]:
```
<keras.src.callbacks.history.History at 0x71ec75cdf320>
```

[12]:
```
class ClusteringLayer(Layer):
    """
```

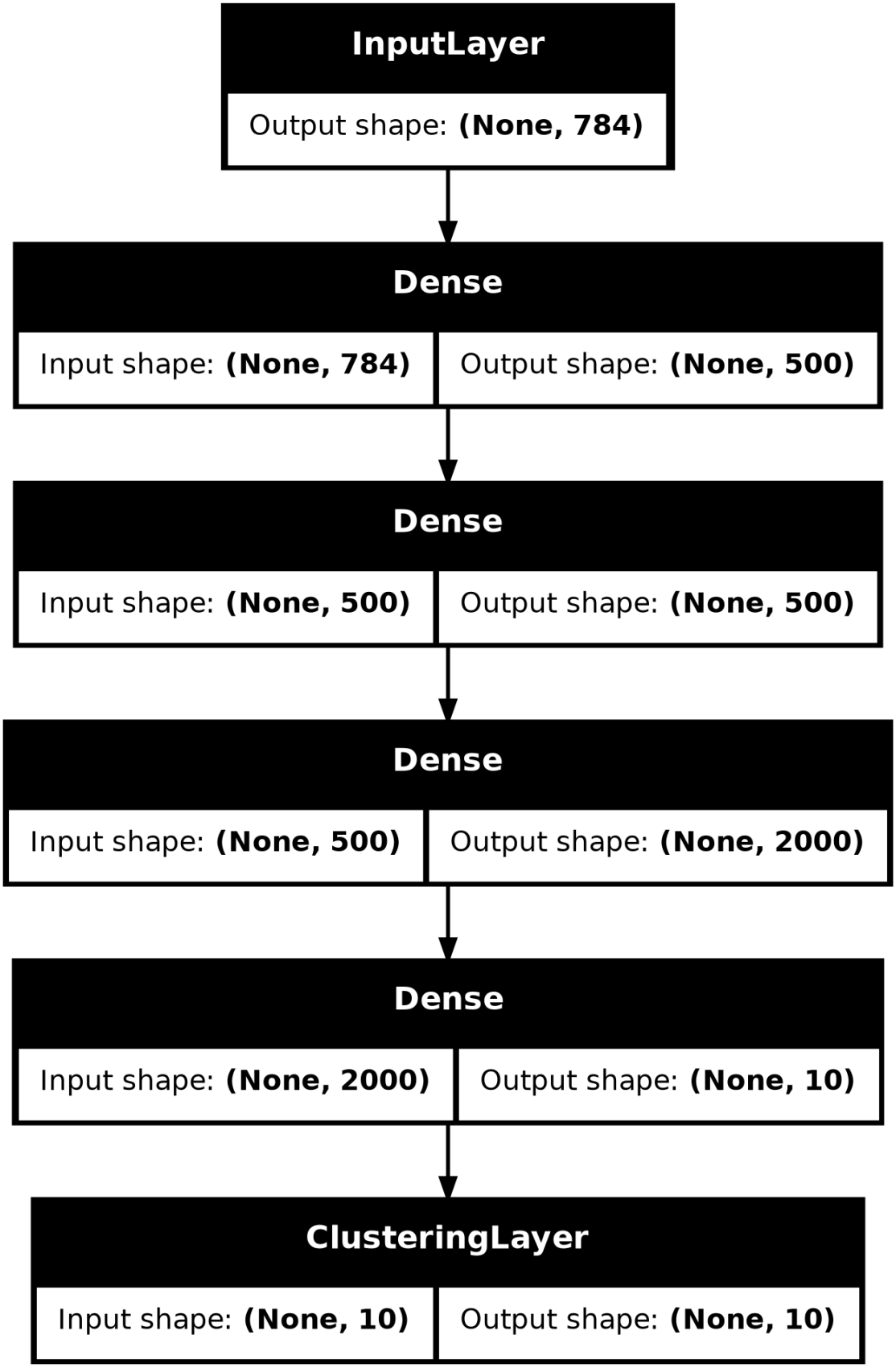
```
    Clustering layer converts input sample (feature) to soft label, i.e. a␣
↪vector that represents the probability of the
    sample belonging to each cluster. The probability is calculated with␣
↪student's t-distribution.

    # Example
    ```
        model.add(ClusteringLayer(n_clusters=10))
    ```
    # Arguments
        n_clusters: number of clusters.
        weights: list of Numpy array with shape `(n_clusters, n_features)`␣
↪witch represents the initial cluster centers.
        alpha: degrees of freedom parameter in Student's t-distribution.␣
↪Default to 1.0.
    # Input shape
        2D tensor with shape: `(n_samples, n_features)`.
    # Output shape
        2D tensor with shape: `(n_samples, n_clusters)`.
    """

    def __init__(self, n_clusters, weights=None, alpha=1.0, **kwargs):
        if 'input_shape' not in kwargs and 'input_dim' in kwargs:
            kwargs['input_shape'] = (kwargs.pop('input_dim'),)
        super(ClusteringLayer, self).__init__(**kwargs)
        self.n_clusters = n_clusters
        self.alpha = alpha
        self.initial_weights = weights
        self.input_spec = InputSpec(ndim=2)

    def build(self, input_shape):
        assert len(input_shape) == 2
        input_dim = input_shape[1]
        self.input_spec = InputSpec(dtype=K.floatx(), shape=(None, input_dim))
       # self.clusters = self.add_weight((self.n_clusters, input_dim),␣
↪initializer='glorot_uniform', name='clusters')
        self.clusters = self.add_weight(shape=(self.n_clusters, input_dim),␣
↪initializer='glorot_uniform', name='clusters')
        if self.initial_weights is not None:
            self.set_weights(self.initial_weights)
            del self.initial_weights
        self.built = True

    def call(self, inputs, **kwargs):
        """ student t-distribution, as same as used in t-SNE algorithm.
         Measure the similarity between embedded point z_i and centroid μ_j.
```

```
                q_ij = 1/(1+dist(x_i, μ_j)^2), then normalize it.
                q_ij can be interpreted as the probability of assigning sample␣
↪i to cluster j.
                (i.e., a soft assignment)
        Arguments:
            inputs: the variable containing data, shape=(n_samples, n_features)
        Return:
            q: student's t-distribution, or soft labels for each sample.␣
↪shape=(n_samples, n_clusters)
        """
        q = 1.0 / (1.0 + (K.sum(K.square(K.expand_dims(inputs, axis=1) - self.
↪clusters), axis=2) / self.alpha))
        q **= (self.alpha + 1.0) / 2.0
        q = K.transpose(K.transpose(q) / K.sum(q, axis=1)) # Make sure each␣
↪sample's 10 values add up to 1.
        return q

    def compute_output_shape(self, input_shape):
        assert input_shape and len(input_shape) == 2
        return input_shape[0], self.n_clusters

    def get_config(self):
        config = {'n_clusters': self.n_clusters}
        base_config = super(ClusteringLayer, self).get_config()
        return dict(list(base_config.items()) + list(config.items()))
```

[13]:
```
clustering_layer = ClusteringLayer(n_clusters, name='clustering')(encoder.
↪output)
model = Model(inputs=encoder.input, outputs=clustering_layer)
```

[14]:
```
from keras.utils import plot_model
plot_model(model, to_file='model.png', show_shapes=True)
from IPython.display import Image
Image(filename='model.png')
```

[14]:


InputLayer
Output shape: (None, 784)
Dense
Input shape: (None, 784)
Output shape: (None, 500)
Dense
Input shape: (None, 500)
Output shape: (None, 500)
Dense
Input shape: (None, 500)
Output shape: (None, 2000)
Dense
Input shape: (None, 2000)
Output shape: (None, 10)
ClusteringLayer
Input shape: (None, 10)
Output shape: (None, 10)

[15]:
```
model.compile(optimizer=SGD(0.01, 0.9), loss='kld')
```

[16]:
```
kmeans = KMeans(n_clusters=n_clusters, n_init=1)
y_pred = kmeans.fit_predict(encoder.predict(x))
```

```
2026-03-10 09:46:04.130581: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_25', 396 bytes spill stores, 432 bytes spill loads

2026-03-10 09:46:04.247035: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_18', 396 bytes spill stores, 432 bytes spill loads

2026-03-10 09:46:04.468136: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_32', 440 bytes spill stores, 440 bytes spill loads

2026-03-10 09:46:04.728363: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_32', 100 bytes spill stores, 100 bytes spill loads


2126/2188              0s 663us/step
2026-03-10 09:46:07.402517: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_18', 208 bytes spill stores, 208 bytes spill loads

2026-03-10 09:46:07.412249: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_25', 208 bytes spill stores, 208 bytes spill loads

2026-03-10 09:46:07.802177: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_32', 100 bytes spill stores, 100 bytes spill loads


2188/2188              5s 1ms/step
```

[17]:
```
y_pred_last = np.copy(y_pred)
```

[18]:
```
model.get_layer(name='clustering').set_weights([kmeans.cluster_centers_])
```

[19]:
```
kmeans.cluster_centers_
```

[19]:
```
array([[-1.2938418 ,  1.6320109 ,  3.0731578 ,  2.8865423 ,  3.0715053 ,
        -0.70037305,  0.4254169 , -0.69009143, -0.35584134, -0.6989224 ],
       [-0.00683951,  0.3791336 ,  0.5834526 ,  0.1553939 , -0.0668086 ,
        -1.2008235 , -0.02439652,  0.19014454,  0.21889937, -0.76639247],
       [ 2.3045719 , -1.5018439 ,  0.8334174 ,  3.318633  ,  1.9441344 ,
        -2.214971  , -0.33049375,  0.3989861 , -0.32695633, -2.036221  ],
       [-1.0239747 ,  0.7057961 ,  0.83023643,  1.103124  , -1.8906974 ,
        -2.7015886 ,  3.231135  , -0.88751143,  0.63559186, -0.5986897 ],
       [ 0.29612875,  1.4006175 ,  1.8795217 ,  0.4214204 ,  0.33446056,
        -4.055977  , -1.6433809 , -0.41450685,  0.01195937, -1.3394488 ],
       [-0.8709988 ,  0.8041334 , -2.6671944 ,  2.1616435 ,  3.0206149 ,
        -0.02666074, -0.6654121 ,  0.35837722, -1.7828312 , -1.1211815 ],
       [ 1.2678897 , -0.29747388, -0.53745186, -1.894831  , -3.3803754 ,
         0.53302497,  0.9084464 ,  0.05539823, -0.24965939,  1.2524886 ],
       [ 1.9666445 ,  0.83614504,  0.40882534, -0.83296996, -1.5019718 ,
        -2.4027386 ,  0.56432784, -1.561851  , -1.758539  , -1.9331484 ],
       [ 0.06316242,  0.21715926,  0.3970744 ,  0.6270318 ,  1.9632155 ,
         2.9790583 ,  0.00728829, -0.7661491 , -0.34148782,  0.5978107 ],
       [ 1.127549  , -0.8659761 , -1.2252767 , -0.12470281, -2.4512334 ,
        -1.1073351 ,  1.5590907 ,  0.11200997, -1.9847163 ,  0.29445398]],
      dtype=float32)
```

[20]:
```
def target_distribution(q):
    weight = q ** 2 / q.sum(0)
    return (weight.T / weight.sum(1)).T
```

[21]:
```
loss = 0
index = 0
maxiter = 3000
update_interval = 140
index_array = np.arange(x.shape[0])
```

[22]:
```
tol = 0.001 # tolerance threshold to stop training
```

[23]:
```
import tensorflow as tf
for ite in range(int(maxiter)):
    if ite % update_interval == 0:
        q = model.predict(x, verbose=0)
        p = target_distribution(q)  # update the auxiliary target distribution p

        # evaluate the clustering performance
```

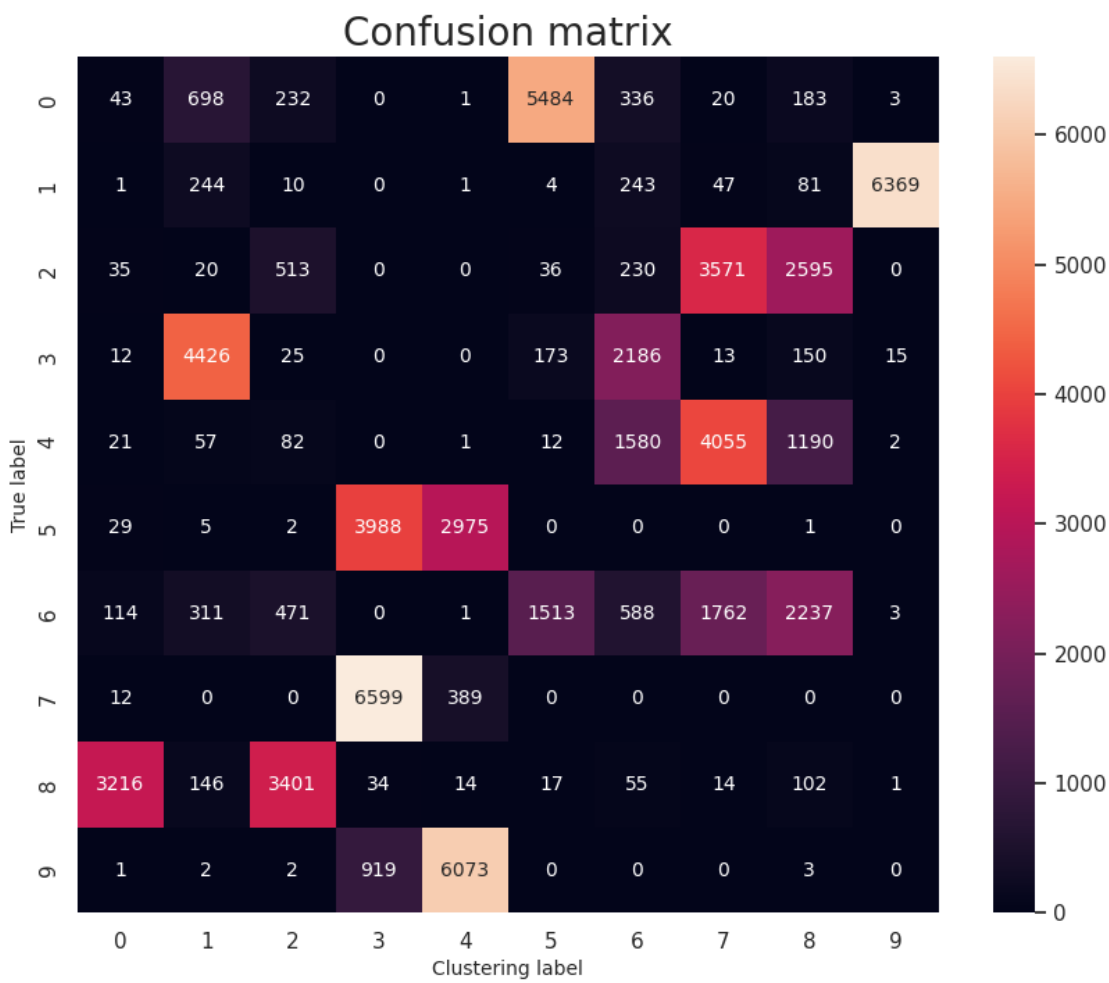
```
        y_pred = q.argmax(1)
        if y is not None:
            acc = np.round(metrics.accuracy_score(y, y_pred), 5)
            nmi = np.round(metrics.normalized_mutual_info_score(y, y_pred), 5)
            ari = np.round(metrics.adjusted_rand_score(y, y_pred), 5)
            loss = np.round(loss, 5)
            print('Iter %d: acc = %.5f, nmi = %.5f, ari = %.5f' % (ite, acc,␣
↪nmi, ari), ' ; loss=', loss)

        # check stop criterion - model convergence
        delta_label = np.sum(y_pred != y_pred_last).astype(np.float32) / y_pred.
↪shape[0]
        y_pred_last = np.copy(y_pred)
        if ite > 0 and delta_label < tol:
            print('delta_label ', delta_label, '< tol ', tol)
            print('Reached tolerance threshold. Stopping training.')
            break
    idx = index_array[index * batch_size: min((index+1) * batch_size, x.
↪shape[0])]
    loss = model.train_on_batch(x=x[idx], y=p[idx])
    index = index + 1 if (index + 1) * batch_size <= x.shape[0] else 0
```

```
Iter 0: acc = 0.09996, nmi = 0.59180, ari = 0.43423  ; loss= 0

2026-03-10 09:46:13.405673: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_58', 184 bytes spill stores, 184 bytes spill loads

2026-03-10 09:46:13.434462: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_58', 84 bytes spill stores, 84 bytes spill loads

2026-03-10 09:46:13.510850: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_58', 408 bytes spill stores, 408 bytes spill loads


Iter 140: acc = 0.10134, nmi = 0.59560, ari = 0.43985  ; loss= 0.1163

2026-03-10 09:46:18.128944: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_58', 136 bytes spill stores, 136 bytes spill loads

2026-03-10 09:46:18.257169: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
```

```
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_58', 84 bytes spill stores, 84 bytes spill loads

2026-03-10 09:46:18.373459: I
external/local_xla/xla/stream_executor/cuda/cuda_asm_compiler.cc:397] ptxas
warning : Registers are spilled to local memory in function
'gemm_fusion_dot_58', 500 bytes spill stores, 396 bytes spill loads


Iter 280: acc = 0.10493, nmi = 0.60476, ari = 0.44707  ; loss= 0.15857
Iter 420: acc = 0.10577, nmi = 0.61358, ari = 0.45647  ; loss= 0.17786
Iter 560: acc = 0.10789, nmi = 0.61950, ari = 0.46244  ; loss= 0.18721
Iter 700: acc = 0.10819, nmi = 0.62544, ari = 0.46983  ; loss= 0.19224
Iter 840: acc = 0.10723, nmi = 0.62783, ari = 0.47250  ; loss= 0.19461
Iter 980: acc = 0.10704, nmi = 0.62975, ari = 0.47474  ; loss= 0.1954
Iter 1120: acc = 0.10770, nmi = 0.63111, ari = 0.47654  ; loss= 0.19543
Iter 1260: acc = 0.10780, nmi = 0.63150, ari = 0.47756  ; loss= 0.19485
Iter 1400: acc = 0.10664, nmi = 0.63163, ari = 0.47807  ; loss= 0.19398
Iter 1540: acc = 0.10534, nmi = 0.63160, ari = 0.47841  ; loss= 0.19288
Iter 1680: acc = 0.10551, nmi = 0.63241, ari = 0.47939  ; loss= 0.19164
Iter 1820: acc = 0.10570, nmi = 0.63254, ari = 0.47948  ; loss= 0.19034
Iter 1960: acc = 0.10649, nmi = 0.63256, ari = 0.47872  ; loss= 0.18893
Iter 2100: acc = 0.10493, nmi = 0.63162, ari = 0.47814  ; loss= 0.18747
Iter 2240: acc = 0.10464, nmi = 0.63146, ari = 0.47829  ; loss= 0.18597
Iter 2380: acc = 0.10409, nmi = 0.63200, ari = 0.47924  ; loss= 0.18454
Iter 2520: acc = 0.10347, nmi = 0.63174, ari = 0.47912  ; loss= 0.18309
Iter 2660: acc = 0.10314, nmi = 0.63175, ari = 0.47887  ; loss= 0.18169
Iter 2800: acc = 0.10254, nmi = 0.63128, ari = 0.47835  ; loss= 0.18027
Iter 2940: acc = 0.10190, nmi = 0.63091, ari = 0.47807  ; loss= 0.17889
```

[24]:
```
# model.trainable_weights
```

[25]:
```
# Eval.
q = model.predict(x, verbose=0)
p = target_distribution(q)  # update the auxiliary target distribution p

# evaluate the clustering performance
y_pred = q.argmax(1)
if y is not None:
    acc = np.round(metrics.accuracy_score(y, y_pred), 5)
    nmi = np.round(metrics.normalized_mutual_info_score(y, y_pred), 5)
    ari = np.round(metrics.adjusted_rand_score(y, y_pred), 5)
    loss = np.round(loss, 5)
    print('Acc = %.5f, nmi = %.5f, ari = %.5f' % (acc, nmi, ari), ' ; loss=',␣
 ↪loss)
```

```
Acc = 0.02130, nmi = 0.62584, ari = 0.47654  ; loss= 0.17212
```

[26]:
```
import seaborn as sns
import sklearn.metrics
import matplotlib.pyplot as plt
sns.set(font_scale=1)
confusion_matrix = sklearn.metrics.confusion_matrix(y, y_pred)

plt.figure(figsize=(10, 8))
sns.heatmap(confusion_matrix, annot=True, fmt="d", annot_kws={"size": 10});
plt.title("Confusion matrix", fontsize=20)
plt.ylabel('True label', fontsize=10)
plt.xlabel('Clustering label', fontsize=10)
plt.show()
```